Subsonus

Subsonus is a next-generation USBL/INS that provides high accuracy position, velocity and heading at depths of up to 1,000 meters.

Position Accuracy	0.1 m
Roll & Pitch	0.1 °
Acoustic Heading	0.3 °
Range & Depth	1,000 m

Get A Quote

Datasheet

Subsonus is a miniature underwater acoustic positioning system that provides high accuracy position, velocity, and heading at ranges of up to 1000 meters. The USBL provides highly reliable tracking thanks to its advanced signal processing and unique hydrophone design. Subsonus also seamlessly operates as a modem capable of transmitting user data underwater.

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Subsonus with USBL2 technology
Advanced Navigation

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Subsonus USBL - How to get accurate underwater
Advanced Navigation

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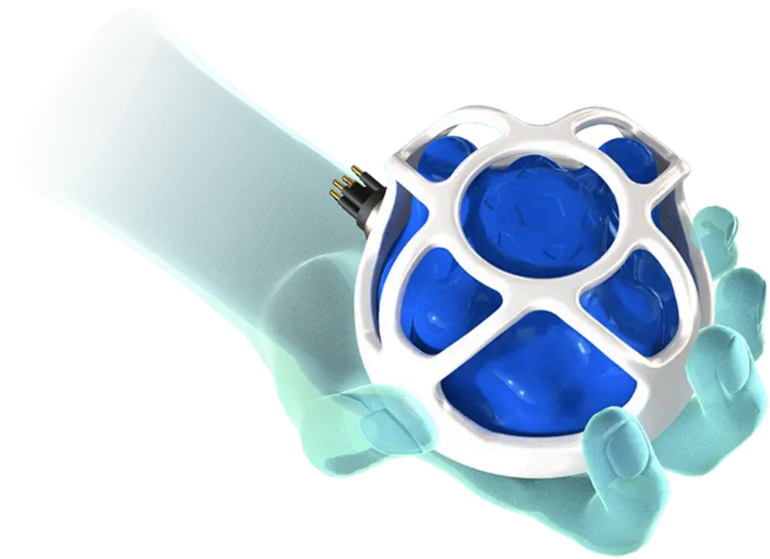
Features Applications Specifications Downloads

Features



Industry Leading Hydrophone Array

Subsonus features an industry-leading eight-channel factory-calibrated hydrophone array. With the hydrophones, Subsonus is able to perform beamforming, offering exceptional multipath rejection in poor environments and higher accuracy measurements.

Dynamic Power and Signal Encoding

Subsonus dynamically adjusts its acoustic transmit power and signal encoding based upon its operating environment. This results in highly improved performance and reliability in difficult conditions.

Acoustic Heading

Subsonus features acoustic heading transfer technology that allows it to transfer high accuracy GNSS heading from the surface to a unit underwater. This allows underwater units to achieve high accuracy heading without a gyrocompass and with no susceptibility to magnetic interference.

Internal Speed of Sound Sensor

Subsonus has the ability to measure the speed of sound through water using a revolutionary new technique. This means that the system is self tuning and no extra equipment or user intervention is required to tune the system for optimal performance.

Fully Integrated Miniature Enclosure

Subsonus does away with the typical reliance on external equipment such as rack mount units, interface boxes or PCs. All processing is done internally inside the miniature titanium enclosure and the system connects through a single ethernet connection for data output. It features a web browser based user interface.

Acoustic Modem

Subsonus offers acoustic modem functionality, capable of rapidly transmitting third-party data via a TCP Network port from one subsonus to another.

Miniaturization

Subsonus utilizes modern high-end electronics and manufacturing technologies to miniaturize the physical components of the USBL system, without compromising on performance. Subsonus runs fully on the processor within the transceiver, and is accessible via a web browser, with no additional rack-mounted interface units or processing PCs.

Applications

AUVs

ROVs

Subsea Surveying

Diver Tracking

Specifications

Navigation

Acoustics

Sensors

Hardware

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Datasheet

Version 3.4 | 28 Apr 2026

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Reference Manual

Online Version

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Version 2.7 | 4 May 2026

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Performance Evaluation

Version 1.1 | 22 Jul 2016

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3D Model

Version 1.0.1 | 5 May 2026

Download

Subsonus Firmware

Version 3.90 | 30 Sep 2025

Download

The latest firmware for all Subsonus hardware versions. This can be loaded onto Subsonus using the web browser interface. Subsonus firmware v3.0 or newer must be used with Subsonus Tag firmware v2.0. Refer to the release notes for further details.

Version 2.4 | 29 Oct 2020

Download

Pre-version 3.0 firmware for all Subsonus hardware versions. This can be loaded onto Subsonus using the web browser interface. Subsonus firmware v2.4 must be used with Subsonus Tag firmware v1.4. Refer to the release notes for further details.

Subsonus SDK

Version 2.4 | 27 Oct 2020

Download

The Subsonus SDK provides example source code for interfacing with Subsonus through the AN Packet Protocol. The languages provided are C/C++, Java, and .Net C#.

Subsonus Tools

Version 3.1 | 5 May 2023

Download

Subsonus Tools is a utility for logging data and converting log files from Subsonus into CSV formats.